

Subhajit Chowdhury

Data Engineer

+91 7029860941 | Kolkata, India | er.subhajitchowdhury@gmail.com
LinkedIn | GitHub | Portfolio | Medium

PROFESSIONAL SUMMARY

Cloud Data Engineer with **4+ years of experience** across **AWS, Snowflake, and dbt**, with working knowledge of **Databricks** (Delta Lake, DLT, Unity Catalog) and **AI/GenAI** (RAG, LLM, AI Agents). Skilled in building enterprise-scale **PySpark ETL/ELT pipelines**, cloud lakehouses, and event-driven architectures for global clients. Proven track record of driving cost savings, sub-minute latency, and end-to-end ownership across distributed IST/EST/GMT delivery.

TECHNICAL SKILLS

Core: PySpark, Python, SQL, ETL/ELT, Spark Tuning, Data Warehousing, Distributed Systems, Real-Time Streaming
Cloud Platforms: AWS (Glue, EMR, S3, Lambda, Step Functions, Athena, CloudWatch, Redshift, DMS, Kinesis, EventBridge, DynamoDB), Snowflake, Snowpipe
Databricks (Working Knowledge): Databricks, Delta Lake, Delta Live Tables, Unity Catalog, Iceberg
Analytics Engineering: dbt (Working Knowledge), Star/Snowflake Schemas, Dimensional Modeling, CDC/SCD Type 1&2
AI & GenAI (Working Knowledge): RAG, LLM Integration, AI Agents, Agentic AI Patterns, Vector Databases
Lakehouse & Modeling: Medallion Architecture (Bronze/Silver/Gold), Metadata-Driven Ingestion, Data Lakehouse
Orchestration: AWS Step Functions, Apache Airflow, EventBridge
DevOps & Collaboration: Jenkins, Git, GitHub Actions, CI/CD, Unix/Linux Shell, Confluence, Jira, Slack, Code Reviews, Data Quality, Integration Testing, UAT, Agile/Scrum
Storage & Formats: Parquet, Delta Lake, Iceberg, JSON, CSV, XML, Avro, PostgreSQL, MongoDB, DynamoDB

PROFESSIONAL EXPERIENCE

Tata Consultancy Services (TCS) – Kolkata, India

Data Engineer, July 2022 – Present

- Built and deployed **200+ production ETL/ELT pipelines** on **AWS Glue (PySpark), EMR, S3, and Athena** processing hundreds of GBs daily under zero data-loss SLA in medallion architecture (Bronze/Silver/Gold) for global clients.
- Migrated legacy on-prem systems (**Oracle, SQL Server, MySQL, MongoDB**) to serverless AWS lakehouses using **AWS DMS, Lambda, Glue, Step Functions, and Athena** – driving approximately **GBP 1.5M+ in yearly infrastructure cost savings** across two enterprise engagements with zero missed milestones.
- Optimised **Spark-based data processing jobs** on **Athena and Redshift** using star schema, dimensional modeling (SCD Type 1/2), Parquet tuning, partition pruning, and predicate pushdown – cutting analytical query costs by **25%** and reducing dashboard load times from **~40s to under 15s** for BI, QA, and Analytics teams.
- Engineered an automated **metadata-driven ingestion framework** with schema discovery, drift detection, lineage tracking, and data quality validation across **50+ datasets** in Parquet, JSON, CSV, and XML formats – cutting downstream data delivery latency by **30%**.
- Architected an event-driven ingestion pipeline using **AWS Lambda, S3 Event Notifications, DynamoDB Streams, Snowpipe, and Snowflake** – dropping data availability lag from **20+ hours to sub-minute** and removing 100% of manual downstream data requests.
- Developed reusable **Python** and **Unix shell** automation integrated with **Jenkins CI/CD, GitHub Actions, and Git** – cutting 40+ weekly deployment interruptions and maintaining cloud production stability via CloudWatch monitoring, root-cause analysis, and SLA-bound incident resolution while collaborating asynchronously with data analysts, BI engineers, and technical architects in Agile/Scrum delivery.

PROJECTS

- Near-Real-Time ELT Pipeline on AWS & Snowflake** – Architected a serverless event flow (**S3 Events** → **Lambda** → **DynamoDB** → **Snowpipe** → **Snowflake**) achieving sub-minute ingestion latency and replacing a 4-hour batch dependency at zero standing infrastructure cost.
- Serverless Medallion Lakehouse on AWS** – Built a Bronze/Silver/Gold medallion lakehouse using **Glue (PySpark), S3, Athena, and Step Functions** enabling analysts to run self-service Athena SQL on curated Parquet datasets at zero standing cluster cost.

ACHIEVEMENTS & CERTIFICATIONS

- Drove approximately **GBP 1.5M+** in annualised infrastructure savings across two enterprise cloud migrations
- Delivered **200+ production data pipelines** under zero data-loss SLA across global enterprise engagements
- AWS Certified Cloud Practitioner** (July 2025 – July 2028) – Verify Credential - a6447414ae25419cba12941481cd10a2
- AWS Certified Data Engineer – Associate (DEA-C01)** – *In Progress*

EDUCATION

Modern Institute of Engineering and Technology, MAKAUT – Kolkata, India

Bachelor of Technology in Computer Science and Engineering, July 2018 – June 2022

Graduated with 8.90/10 CGPA